



Irritable Bowel Syndrome (IBS)

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Irritable bowel syndrome is characterized by recurrent abdominal discomfort or pain with at least two of the following characteristics: relation to defecation, association with a change in frequency of stool, or association with a change in consistency of stool. The cause is not known, and the pathophysiology is incompletely understood. Diagnosis is clinical. Treatment is symptomatic, consisting of dietary management and medications, including anticholinergics and agents active at serotonin receptors.

Etiology of IBS

Irritable bowel syndrome (IBS—previously called functional gastrointestinal [GI] disorders) is a disorder of gut-brain interaction. No anatomic cause can be found on laboratory tests, imaging studies, and biopsies.

Emotional factors, diet, viral illnesses, or hormones may precipitate or aggravate GI symptoms.

Historically, the disorder was often considered as purely psychosomatic. Although psychosocial factors can be involved, IBS is better understood as a combination of physiologic and psychosocial factors.

Physiologic factors

A variety of physiologic factors seem to be involved in IBS symptoms. These factors include

- Increased intestinal sensitivity (visceral hyperalgesia)
- Altered intestinal motility

Visceral hyperalgesia refers to hypersensitivity to normal amounts of intraluminal distention and heightened perception of pain in the presence of normal quantities of intestinal gas; it may result from remodeling of neural pathways in the gut-brain axis. Some patients (perhaps 1 in 7) have reported their IBS symptoms began after an episode of **acute gastroenteritis** (termed postinfectious IBS). However, many patients have no demonstrable physiologic abnormalities, and, even in those who do, the abnormalities may not correlate with symptoms.

Constipation may be explained by slower colonic transit, and diarrhea may be explained by faster colonic transit. Some patients with constipation have fewer colonic high amplitude-propagated contractions, which propel colonic contents over several segments.

Postprandial abdominal discomfort may be attributed to an exaggerated gastro-colonic reflex (the colonic contractile response to a meal), the presence of colonic high amplitude-propagated contractions, visceral hyperalgesia, or a combination of these factors. Fat ingestion may increase intestinal permeability and exaggerate hypersensitivity. Foods high in fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (collectively called FODMAPs) are poorly absorbed in the small intestine and may increase colonic motility and secretion.

Hormonal fluctuations affect bowel functions in women. Rectal sensitivity is increased during menses but not during other phases of the menstrual cycle. The effects of sex hormones on gastrointestinal transit are subtle.

Psychosocial factors

Psychological distress is common among patients with IBS, especially in those who seek medical care. Some patients have anxiety disorders, depression, or a somatization disorder. Sleep disturbances also coexist. However, stress and emotional conflict do not always coincide with symptom onset and recurrence. Some patients with IBS seem to have a learned aberrant illness behavior (ie, they express emotional conflict as a gastrointestinal complaint, usually abdominal pain). The physician evaluating patients with IBS, particularly those with refractory symptoms, should investigate for unresolved psychological issues, including the possibility of sexual or physical abuse. Psychosocial factors can also affect the outcome in IBS.

Symptoms and Signs of IBS

Irritable bowel syndrome tends to begin in adolescence and the 20s, causing bouts of symptoms that recur at irregular periods. Onset in late adult life is also possible but less common. Symptoms of IBS rarely rouse the sleeping patient. Symptoms are often triggered by food or by stress.

Patients have abdominal discomfort, which varies considerably but is often located in the lower abdomen, is steady or cramping in nature, and is related to defecation. In addition, abdominal discomfort is temporally associated with alterations in stool frequency (increased in diarrhea-predominant IBS and decreased in constipation-predominant IBS) and consistency (ie, loose or lumpy and hard). Pain or discomfort related to defecation is likely to be of bowel origin; that associated with exercise, movement, urination, or menstruation usually has a different cause.

Although bowel patterns are relatively consistent in most patients, it is not unusual for patients to alternate between constipation and diarrhea. Patients may also have symptoms of abnormal stool passage (straining, urgency, or feeling of incomplete evacuation), pass mucus, or complain of bloating or abdominal distention. Many patients also have symptoms of dyspepsia. Extraintestinal symptoms (eg, fatigue, fibromyalgia, sleep disturbances, chronic headaches) are common.

Diagnosis of IBS

- Clinical evaluation, based on Rome criteria
- Limited laboratory tests
- Other tests for patients with red flag findings

Diagnosis of irritable bowel syndrome is based on history, specifically characteristic bowel patterns, time and character of pain, and absence of red flags, and on a focused physical examination.

Red flags

Diagnostic testing should be more intensive when the following red flags are present either at initial presentation or at any time after diagnosis:

- Older age
- Weight loss
- Rectal bleeding
- Iron deficiency anemia
- Family history of colon cancer, inflammatory bowel disease, or celiac disease
- Nocturnal diarrhea

Differential diagnosis

Because patients with IBS can develop organic conditions, testing for other conditions should also be considered in patients who have red flags or develop markedly different symptoms during the course of IBS. Common illnesses that may be confused with IBS include

- [Lactose intolerance](#)
- [Medication-induced diarrhea](#)
- [Postcholecystectomy syndrome](#)
- [Laxative abuse](#)
- Parasitic diseases (eg, [giardiasis](#))
- [Eosinophilic gastritis](#) or enteritis
- Microscopic colitis
- [Small intestinal bacterial overgrowth](#)
- [Celiac disease](#)
- Bile acid diarrhea
- Early [inflammatory bowel disease](#)

However, uninflamed colonic diverticula do not cause symptoms, and their presence should not be considered explanatory.

The bimodal age distribution of patients with inflammatory bowel disease makes it imperative to evaluate both younger and older patients to rule out inflammatory bowel disease as a cause of the patient's symptoms. In patients > age 60 with acute symptoms, [ischemic colitis](#) should be considered. Patients with constipation and no anatomic lesions should be evaluated for [hypothyroidism](#) and [hypercalcemia](#). If the patient's symptoms suggest [malabsorption](#) or [celiac disease](#), testing should be done. Defecatory disorders should be considered as a cause of constipation in patients who report symptoms of difficult defecation.

Rare causes of diarrhea include [hyperthyroidism](#), [medullary cancer of the thyroid](#), [carcinoid syndrome](#), [gastrinoma](#), [vipoma](#), [tropical sprue](#), and [Whipple disease](#).

Pearls & Pitfalls

- Uninflamed colonic diverticula do not cause symptoms, so their presence does not explain gastrointestinal symptoms.

History

Particular attention should be given to the character of the pain, bowel habits, familial interrelationships, medication, and dietary histories. Equally important are the patient's overall emotional state, interpretation of personal problems, and quality of life. The quality of the patient-physician interaction is key to diagnostic and therapeutic efficacy.

The **Rome criteria** are standardized symptom-based criteria for diagnosing IBS. The Rome criteria require the presence of abdominal pain for at least 1 day/week in the last 3 months along with ≥ 2 of the following (1):

- Pain is related to defecation.
- Pain is associated with a change in frequency of defecation.
- Pain is associated with a change in consistency of stool.

Physical examination

Patients generally appear to be healthy.

Palpation of the abdomen may reveal tenderness, particularly in the left lower quadrant, at times associated with a palpable, tender sigmoid.

A digital rectal examination, including a test for occult blood, should be done on all patients.

In women, a [pelvic examination](#) helps rule out ovarian tumors and cysts or [endometriosis](#), which may mimic IBS.

Testing

(See also the American College of Gastroenterology's [2021 guidelines on the management of irritable bowel syndrome](#).)

The diagnosis of IBS can reasonably be made using the Rome criteria as long as patients have no red flag findings, such as rectal bleeding, weight loss, or other findings that might suggest another etiology. Laboratory testing may include complete blood count and biochemical profile (including liver tests). For patients with diarrhea predominance, serologic markers for [celiac disease](#) (tissue-transglutaminase IgA with an IgA level) and testing for [inflammatory bowel disease](#) with fecal calprotectin or fecal lactoferrin and C-reactive protein are recommended ([1](#)). For patients with constipation, measurement of thyroid-stimulating hormone and calcium levels is recommended.

Testing for enteric pathogens, including *Giardia*, is no longer recommended for patients with IBS unless there is a high pretest probability of infection. If there are definite risk factors for *Giardia* exposure (eg, compromised water supply, travel, daycare setting, camping), fecal immunoassays or polymerase chain reaction testing for *Giardia* is recommended.

Colonoscopy is recommended for patients > age 45 to exclude colonic polyps and tumors. The mucosal and vascular patterns in IBS usually appear normal. In patients with chronic diarrhea (≥ 3 loose stools a day for at least 4 weeks), particularly older women with more severe diarrhea, mucosal biopsy can rule out possible microscopic colitis.

Additional studies (such as ultrasonography, CT, barium enema x-ray, upper gastrointestinal esophagogastroduodenoscopy, and small-bowel x-rays) should be undertaken only when there are other objective abnormalities. Fecal fat excretion or pancreatic elastase should be measured when there is a concern about steatorrhea. Small-bowel evaluation (eg, enteroscopy, capsule endoscopy) is recommended when [malabsorption](#) is suspected. Testing for [carbohydrate intolerance](#) or [small intestinal bacterial overgrowth](#) should be considered in appropriate circumstances.

Intercurrent disease

Patients with IBS may subsequently develop additional gastrointestinal disorders, and the clinician must not summarily dismiss their complaints. Changes in symptoms (eg, in the location, type, or intensity of pain; in bowel habits; in constipation and diarrhea) and new symptoms or complaints (eg, nocturnal diarrhea) may signal another disease process.

Other symptoms that require investigation include fresh blood in the stool, weight loss, very severe abdominal pain or unusual abdominal distention, steatorrhea or noticeably foul-smelling stools, fever or chills, persistent vomiting, hematemesis, symptoms that wake the patient from sleep (eg, pain, the urge to defecate, fecal incontinence), and a steady progressive worsening of symptoms. Patients > age 45 are more likely than younger patients to develop an intercurrent physiologic illness; thus, the threshold for testing should be lower.

Diagnosis reference

1. [Lacy BE, Pimentel M, Brenner DM, et al](#): ACG Clinical Guideline: Management of irritable bowel syndrome. *Am J Gastroenterol* 116(1):17–44, 2021. doi: 10.14309/ajg.0000000000001036

Treatment of IBS

- Support and understanding
- Normal diet, avoiding gas-producing and diarrhea-producing foods
- Increased fiber intake and hydration for constipation
- Medications directed at the dominant symptoms

Therapy is directed at specific symptoms. Patients should be educated about the disorder (eg, normal bowel physiology and the bowel's hypersensitivity to stress and food) and reassured, after appropriate tests, about the absence of a serious or life-threatening disease.

Regular physical activity helps relieve stress and assists in bowel function, particularly in patients with constipation.

(See also the American College of Gastroenterology's [2018 monograph on management of irritable bowel syndrome](#).)

Diet

In general, a normal diet can be followed. Meals should not be overly large, and eating should be slow and paced. Patients with abdominal distention and increased flatulence may benefit from reducing or eliminating beans, cabbage, and other foods containing fermentable carbohydrates. Reduced intake of sweeteners (eg, sorbitol, mannitol, fructose), which are constituents of natural and processed foods (eg, apple and grape juices, bananas, nuts, raisins), may alleviate flatulence, bloating, and diarrhea. Patients with evidence of lactose intolerance should reduce their intake of milk and dairy products. Patients can try reducing their intake of the aforementioned food categories one at a time and noting the effect on their symptoms, or they can try a [low-FODMAP](#) (fermentable oligosaccharides, disaccharides, monosaccharides, and polyols) diet, which restricts all of the aforementioned food categories. If improvement is noted, one food category is added back at a time to see whether there are changes in symptoms ([1](#)). In addition, a low-fat diet may reduce postprandial abdominal symptoms.

Patients should be encouraged to drink more fluids. Dietary soluble fiber supplements may soften stool and improve the ease of evacuation. However, excessive use of fiber can lead to bloating and diarrhea, so fiber doses must be individualized. Occasionally, flatulence may be reduced by switching to a synthetic fiber preparation (eg, [methylcellulose](#), [psyllium](#)).

Medications

(See also the American Gastroenterological Association's [2022 guideline on the pharmacologic management of IBS with diarrhea](#) and the [2022 guideline on the pharmacologic management of IBS with constipation](#).)

Medications are directed toward the dominant symptoms.

Anticholinergic medications (eg, [hyoscyamine](#) 0.125 mg orally 30 to 60 minutes before meals) may be used for their antispasmodic effects, but data on their effectiveness are limited.

In **patients with constipation-predominant IBS (IBS-C)**, the chloride channel activator [lubiprostone](#) and the guanylate cyclase C agonists [linaclotide](#) and [plecanatide](#) may be helpful. [Tenapanor](#) inhibits the gastrointestinal sodium/hydrogen exchange and is available for the treatment of IBS-C. [Polyethylene glycol](#) laxatives have not been well-studied in IBS. However, they have been shown to be effective for use in chronic constipation and for bowel lavage before colonoscopy and are thus frequently used for IBS-C. [Prucalopride](#) is a highly selective serotonin receptor agonist that is available for chronic constipation.

In **patients with diarrhea-predominant IBS (IBS-D)**, diphenoxylate 5 mg/[atropine](#) sulfate or [loperamide](#) may be given before meals. The dose of loperamide should be titrated upward to reduce diarrhea while avoiding constipation (maximum dose 16 mg/day). [Rifaximin](#) is an antibiotic that has been shown to relieve symptoms of bloating and abdominal pain and to help decrease looseness of stools in patients with IBS-D. [Alosetron](#) is a 5-hydroxytryptamine (serotonin) 3 (5HT₃) receptor antagonist that may benefit women with severe IBS-D refractory to other medications. Because alosetron has been associated with ischemic colitis, its use in the United States is under a restricted prescribing program. [Eluxadoline](#) has mixed opioid receptor activity and is indicated for treatment of IBS-D; however, because of the risk of pancreatitis, it cannot be used in patients who have had a cholecystectomy, have sphincter of Oddi dysfunction, have liver disease, or drink more than 3 alcoholic drinks a day.

For many patients, **tricyclic antidepressants (TCAs)** help relieve symptoms of diarrhea, abdominal pain, and bloating. These medications are thought to reduce pain by down-regulating the activity of spinal cord and cortical afferent pathways arriving from the intestine. Secondary amine TCAs (eg, [nortriptyline](#), [desipramine](#)) are often better tolerated than parent tertiary amines (eg, [amitriptyline](#), [imipramine](#), [doxepin](#)) because of fewer anticholinergic, sedating antihistaminic, and alpha-adrenergic adverse effects. Treatment should begin with a very low dose of a TCA (eg, [desipramine](#) 10 to 25 mg orally once a day at bedtime), increasing as necessary and tolerated up to about 200 mg orally once a day.

Selective serotonin reuptake inhibitors are sometimes used in patients with anxiety or an affective disorder, but studies have not shown a significant benefit for patients with IBS and they may exacerbate diarrhea.

The use of **probiotics** to treat IBS has increased in recent years given the importance of the intestinal microbiome in this disorder. However, the data on their effectiveness in treating IBS are limited. [Fecal microbiota](#) transplantation for the treatment of IBS has been studied, but data are limited and inconclusive (2).

Certain **aromatic oils** (carminatives) can relax smooth muscle and relieve pain caused by cramps in some patients. Peppermint oil is the most commonly used agent in this class.

Psychological therapies

Cognitive-behavioral therapy, standard psychotherapy, and [hypnotherapy](#) may help some patients with IBS.

Treatment references

1. [Lacy BE, Pimentel M, Brenner DM, et al](#): ACG Clinical Guideline: Management of irritable bowel syndrome. *Am J Gastroenterol* 116(1):17–44, 2021. doi: 10.14309/ajg.0000000000001036
2. [Wu J, Lv L, Wang C](#): Efficacy of [Fecal Microbiota](#) Transplantation in Irritable Bowel Syndrome: A Meta-Analysis of Randomized Controlled Trials. *Front Cell Infect Microbiol* 12:827395, 2022. doi: 10.3389/fcimb.2022.827395

Key Points

- IBS is recurrent abdominal discomfort or pain accompanied by ≥ 2 of the following: pain related to defecation, pain associated with a change in frequency of stool (diarrhea or constipation), or pain associated with a change in consistency of stool.
- Etiology is unclear but appears to involve both physiologic and psychosocial factors.
- In patients with red flag findings, such as older age, weight loss, or rectal bleeding, more dangerous disorders should be excluded.
- Common illnesses that may be confused with IBS include lactose intolerance, medication-induced diarrhea, post-cholecystectomy diarrhea, laxative abuse, parasitic diseases, eosinophilic gastritis or enteritis, microscopic colitis, small intestinal bacterial overgrowth, celiac disease, and early inflammatory bowel disease.
- Typical testing to consider includes complete blood count, biochemical profile (including liver tests), serologic markers for celiac disease and tests for inflammation (for patients with diarrhea predominance), and measurement of thyroid-stimulating hormone and calcium levels (for patients with constipation).
- A supportive, understanding, and therapeutic physician–patient relationship is essential; direct medications toward the dominant symptoms.

More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

American College of Gastroenterology's [Guidelines on the management of irritable bowel syndrome](#) (2021)

American College of Gastroenterology: [Monograph on management of irritable bowel syndrome](#) (2018)

American Gastroenterological Association: [Clinical practice guideline on the pharmacological management of irritable bowel syndrome with diarrhea](#) (2022)

American Gastroenterological Association: [Clinical practice guideline on the pharmacological management of irritable bowel syndrome with constipation](#) (2022)

National Institute of Diabetes and Digestive and Kidney Diseases: [Useful dietary advice for patients about eating, diet, and nutrition for IBS](#)

American College of Gastroenterology: [Low-FODMAP \(Fermentable, Oligo-, Di-, Mono-saccharides And Polyols\) Diet: Overview](#) (2021)

Drug Information for the Topic



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